

motor, provided their stator has the same design and the same variable frequency drive is used for control. In this case, permanent magnet synchronous electric motors in comparison with other electric motors have the best performance: power/volume, torque/inertia, etc.

A permanent magnet synchronous motor, like any rotating electric motor, consists of a rotor and a stator. The permanent magnet synchronous motor is efficient, brushless, fast, safe, and give high dynamic performance as compared to the conventional motors. It produces smooth torque, low noise and is mainly used for high-speed applications like robotics. It is a 3-phase AC synchronous motor that runs at synchronous speed with the applied AC source.

Instead of using winding for the rotor, permanent magnets are mounted to create a rotating magnetic field. As there is no supply of DC source, this type of motor is very simple and costs less. It contains a stator with three windings installed on it and a rotor with a permanent magnet mounted to create field poles. The 3-phase input AC supply is given to the stator to start working.

The permanent magnet synchronous motor working principle is similar to the synchronous motor. It depends on the rotating magnetic field that generates electromotive force at synchronous speed. When the stator winding is energised by giving 3-phase supply, a rotating magnetic field is created in between the air gaps. This produces the torque when the rotor field poles hold the rotating magnetic field at synchronous speed and the rotor rotates continuously.

As these are not self-starting motors, it is necessary to provide a variable frequency power supply.

Their advantages:

- Provide higher efficiency at high speeds
- Available in small sizes and different packages
- Maintenance and installation is much easier than an induction motor
- Capable of maintaining full torque at low speeds
- High efficiency and reliability
- Give smooth torque and dynamic performance

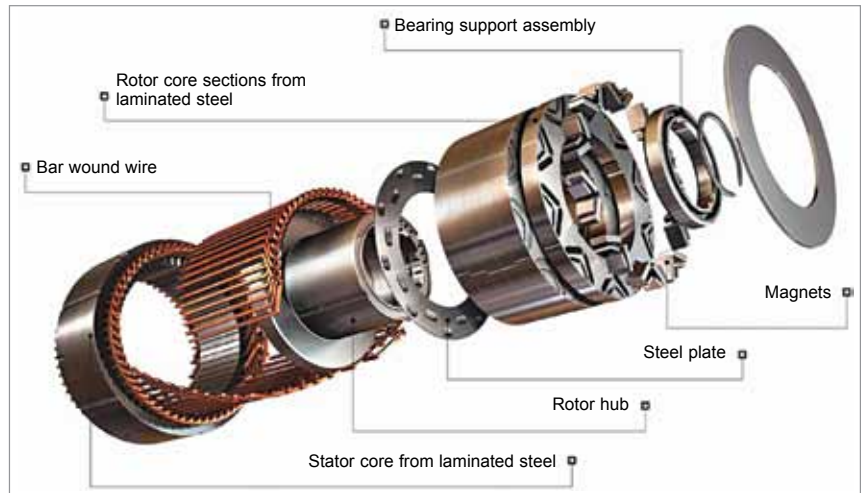


Fig. 4: A permanent magnet synchronous motor

Their disadvantages:

- Very expensive as compared to induction motors
- Somehow difficult to start as these are not self-starting motors

Their applications:

- Air-conditioners and their compressors
- Refrigerators
- Washing machines, which are direct-drive
- Automotive electrical power steering
- Machine tools
- Large power systems to improve leading and lagging power factor
- Control of traction
- Data storage units
- Servo drives
- Industrial applications like robotics, aerospace, and more

Synchronous reluctance motors

Modern synchronous reluctance motors are always inverter powered and use the same type of distributed wound stators as the AC induction motors. However, their rotors are made of laminated steel with punched slots, so these can be easily magnetised in one axis and less so in the other. The tendency of these rotors to cope with the rotating field of the stator creates torque. Like the permanent magnet motors, these are self-synchronous, which means that the inverter synchronises the stator supply to the rotor angle and speed.

The main advantage of synchronous reluctance motors is their negligi-



Fig. 5: A synchronous reluctance motor

ble rotor losses compared to induction motors. So, with thorough design and control, the motors will be able to meet motor efficiency guidelines in future European IE4 and NEMA standards without the need for magnets. The reduced heat in synchronous reluctance motors improves torque and power densities compared to industrial motors and allows them to be usually one dimension smaller for a given rating. These motors are also quite due to low torque ripple and vibration levels.

A somewhat discussed disadvantage of synchronous reluctance motors is their low power factor compared to industrial motors. So, a significantly higher inverter current is required for a given level of mechanical power, which increases the cost and power loss of the inverter. It means, while the efficiency of synchronous reluctance motors can be good, their system-level benefits are less convincing.

Standards currently address efficiency with motors only, and designers should carefully examine the likely overall performance of the